# Example exercise 3 bis

Wednesday, 29 April 2020 11.29

Determine whether the following two CCS expressions

$$a.(b.Nil + c.Nil) \quad \text{and} \quad a.(b.Nil + \tau.(b.Nil + c.Nil))$$

are:

- strongly bisimilar?
- weakly bisimilar?

Solution: The processes above are not strongly bisimilar but they are weakly bisimilar.

Let

$A = a.(b.Nil + c.Nil)$ and

$B = a.(b.Nil + \tau.(b.Nil + c.Nil))$

The LTS generated by A and B are as follows:

$$\begin{tikzcd} A \arrow[r, "a"] & b.Nil + c.Nil \arrow[d, bend right, "b"'] \arrow[d, bend left, "c"] \\ & Nil \end{tikzcd}$$

$$\begin{tikzcd} B \arrow[r, "a"] & b.Nil + \tau.(b.Nil + c.Nil) \arrow[dl, "b"'] \arrow[dr, "\tau"] & \\ Nil & & b.Nil + c.Nil \arrow[ll, bend right, "b"'] \arrow[ll, bend left, "c"] \end{tikzcd}$$

- We can prove that $A \not\sim B$ by providing a distinguishing formula:

  $A \not\models \langle a \rangle \langle \tau \rangle tt$

  $B \models \langle a \rangle \langle \tau \rangle tt$

Then, by Hennessy-Milner theorem, we know that $A \not\sim B$.

- We can prove $A \approx B$ by providing a weak bisimulation relation $R$ containing the pair $(A,B)$.

$$R = \{ (A,B), (b.Nil + c.Nil\ ,\ b.Nil + \tau.(b.Nil + c.Nil)),$$
$$(b.Nil + c.Nil\ ,\ b.Nil + c.Nil),\ (Nil, Nil) \}$$

The relation above is a bisimulation relation because the following conditions hold:

* if $A \xrightarrow{a} b.Nil + c.Nil$ , then $B \xRightarrow{a} b.Nil + c.Nil$ and $(b.Nil + c.Nil, b.Nil + c.Nil) \in R$
* if $(b.Nil + c.Nil) \xrightarrow{b,c} Nil$ , then
  $(b.Nil + \tau.(b.Nil + c.Nil)) \xRightarrow{b,c} Nil$
  and $(Nil, Nil) \in R$
* the last two pairs are clear

And, vice versa

* if $B \xrightarrow{a} b.Nil + \tau.(b.Nil + c.Nil)$
  then $A \xRightarrow{a} b.Nil + c.Nil$ , and
  $(b.Nil + c.Nil\ ,\ b.Nil + \tau.(b.Nil + c.Nil)) \in R$
* If $b.Nil + \tau.(b.Nil + c.Nil) \xrightarrow{\tau} b.Nil + c.Nil$
  then $b.Nil + c.Nil \xRightarrow{\tau} b.Nil + c.Nil$ and
  $(b.Nil + c.Nil\ ,\ b.Nil + c.Nil) \in R$.
* If $b.Nil + \tau.(b.Nil + c.Nil) \xrightarrow{b} Nil$ then
  $b.Nil + c.Nil \xRightarrow{b} Nil$ and $(Nil, Nil) \in R$
* the last two pairs are clear.

Remark: notice that we could have proven $A \approx B$ by providing a universal winning strategy for defender in a weak bisimulation game.